

Topic: Tabular data

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1. Introduction

This topic will teach you about common graphics for showing tabular data. These are the kind of graphics that are common in statistics.

In this book, we draw on material in the public domain, including journal articles and quite a few videos. We also have some interviews with data visualisation experts.



Exploring the learning materials

As you work through the learning materials in this Moodle book, please complete the learning activities and consider the reflection opportunities presented. Various videos and webpages may be included for your reference to reinforce and expand on the contents of this book. You are recommended to watch any videos produced by Monash University. Unless stated otherwise, the other resources are optional but you may find them very helpful. If any links no longer work, try searching for the title elsewhere.



Recommended readings

- Please take a look at the graphics used to visualise multivariate medical data in the video by Agustin Calatroni [Visualizing Multivariate Data: Turning Information Into Understanding](#). It's quite long (1hr 26m), so you can skip some of it.
- [Chapter 7: Arrange Tables of Visualization Analysis and Design](#). Munzner, Tamara. CRC Press, 2014.

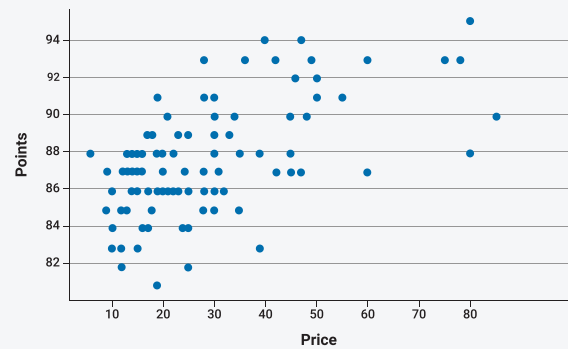
2. Basic statistical graphics

A wide variety of simple statistical graphics are used to show two-dimensional tabular data. These graphics show data distributions and counts of a single attribute or how one attribute changes with respect to another, including time. Such graphics show only the values of a single attribute or the relationship between two attributes. We provide a brief review of the most common.

With tabular data it is common for there to be a *key* (aka *factor*) which is an independent attribute that acts as an index into the data, while a dependent attribute is called a *value*. Key attributes can be categorical or ordinal and may be hierarchically organised. The different values for key attributes are usually called *levels*. Value attributes can be any kind of attribute: categorical, ordinal or quantitative. The most common graphics are as follows:

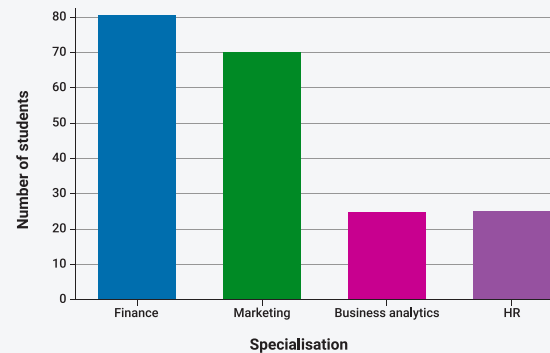
Scatter plots

Scatter plots are one of the most widely used statistical graphics. They show two quantitative value variables and can reveal linear or curvilinear relationships between the variables, correlations between the variables and the presence of extreme values (outliers).



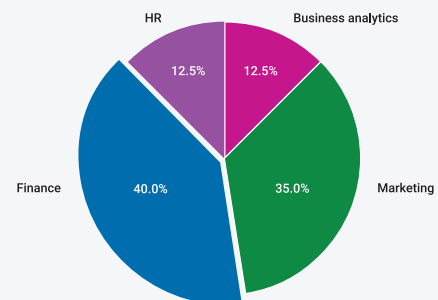
Bar charts

Bar charts and *polar area charts* show a quantitative value for a key. Bar charts are easier to read.



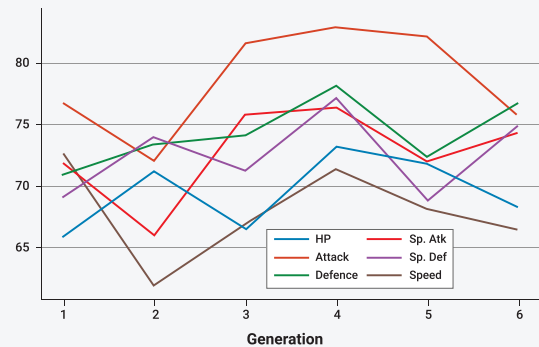
Pie charts

Pie charts or a *normalised stacked bar chart* show relative contribution of parts to a whole. In general pie charts and other radial layouts are harder to read than rectilinear layouts.



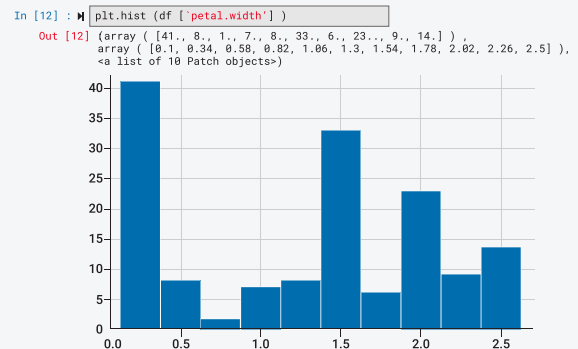
Line charts

Line charts or *dot charts* show one quantitative value for an ordered key attribute. Line charts are used if it makes sense to interpolate between the values. Commonly used to show time varying data.



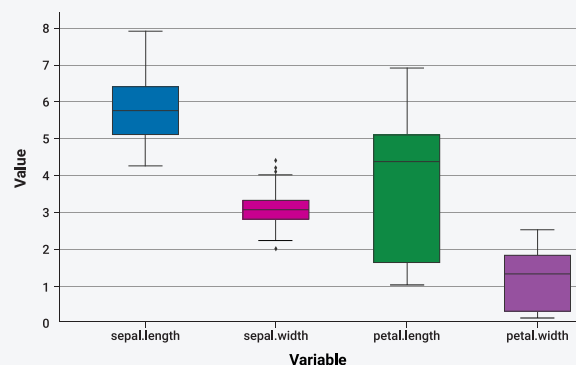
Histograms

Histograms (like a bar chart) and *density plots* (like a line chart) are commonly used to show the frequency of data values for a single attribute.



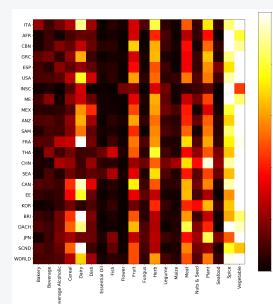
Box-and-whisker plots

Box-and-whisker plots or *violin plots* provide a compact summary of a frequency distribution.



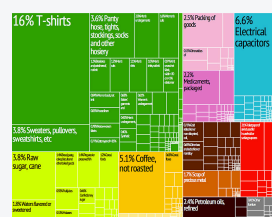
Heat maps

Heat maps are commonly used to show a single value attribute with two keys. The rows or columns can be reordered and clustered so that keys with similar data values are adjacent.



Tree maps

Tree maps can be used to show quantitative data for hierarchically organised keys.



2.1. Making sense of tabular data

Tabular data is a particular type of that can be organised into tables. Rows and columns represent the data items and the attributes of the data.

Watch the following video which discusses the various types of data in tabular data and some history of visualising tabular data. The video also covers univariate and bivariate tabular visualisations and identifying independent versus dependent variables.

Select the advanced player features ^ control to access the video table of contents with chapters targeting each topic.

These captions have been automatically generated. As such, the quality is not 100% accurate. Accuracy may be affected by audio quality, topic under discussion and clarity of the speakers. Staff are not expected or required to edit or correct automatically-generated captions.

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Making sense of tabular data (2022) Monash University <[Panopto Viewer](#)>

2.2. Simple graph visualisations

Purpose

This task is intended to help you to check your understanding of univariate and bivariate data visualisation types.

Task

Step 1: Complete the following activities for each of the following data visualisations:

- The first question asks you to distinguish between **univariate** and **bivariate** data visualisations.
- The next activity asks you to distinguish between the **dependent** and the **independent** variable.

Use the   controls to advance to the next visualisation type. Use the  control to expand to full-screen view.

Check your responses using the  **Check** button.



1 / 7

Bar chart

Is the following data visualisation useful for **univariate** or **bivariate** data?

☐ Bivariate ☐ Univariate

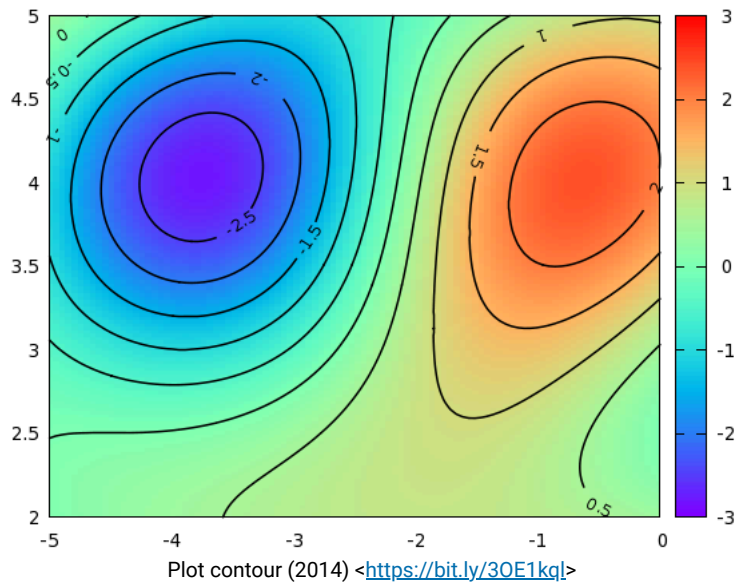
3. Multivariate data

In most real-world data science applications, tabular data is multivariate as it has more than two attributes, often many more. So the question is, how can we show more than two attributes on a two- (or at most three-dimensional) surface? Edward Tufte calls this the problem of leaving flatland. Fortunately, many ingenious graphics for showing multivariate data have been invented and are an important part of the data scientist's visualisation repertoire.

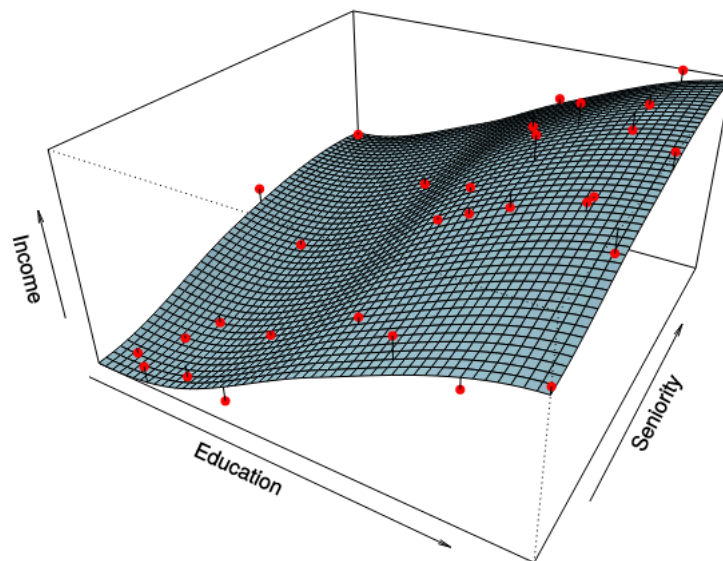
3.1. Showing three dimensions of data

The easiest case is when we add only a single dimension to our problem and wish to visualise the values of three variables. In this case we can use techniques developed in scientific visualisation for showing physical objects such as buildings, crystals or living creatures or the behaviour of fluids, fields or gasses in our three-dimensional world.

One way is to show contours relating to the values of a third variable on a 2D surface:



Another is to extend common 2D graphics with a third dimension and use 3D graphics formats like WebGL to present the resulting 3D objects, point clouds or surfaces in 3D. It is common to plot data points in a 3D scatter plot and show fitted curves using wire mesh surfaces for example.



3D plot of models (2014) <<https://bit.ly/3ik3vDI>>

We can also use animation to show the extra dimension, displaying how the two-dimensional graphic changes at different points in time. This makes most sense when the extra dimension to be visualised is actually time.

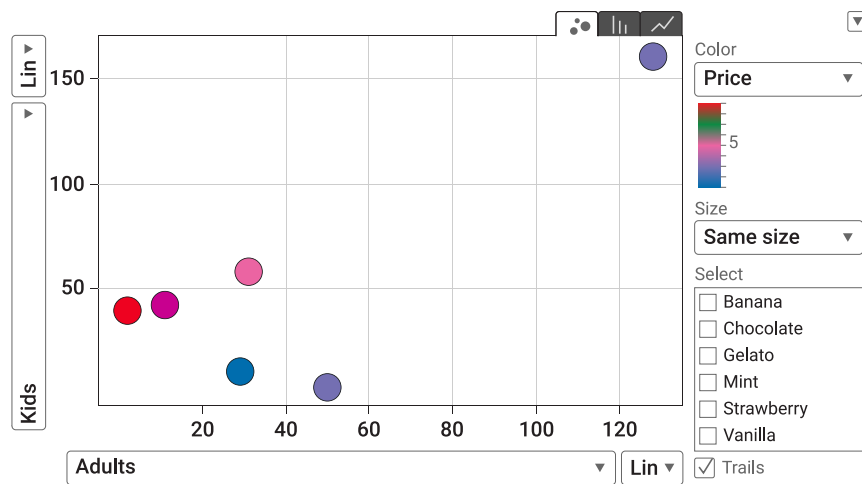
Of course, any of the following, more general, techniques for multivariate visualisation can be used for the particular case of three variables.

3.2. Additional visual attributes

Graphs and charts are made up of different kinds of geometric objects: points, lines, and polygons. These have a location, a shape (i.e., a point might be a cross or a circle, a polygon or a bar), a size or sizes, and visual but non-spatial features such as colour, texture, shadow, line thickness and style, fill colour, blurring, etc. Some of these features such as colour or texture can be broken down even further. For instance, hue, saturation and brightness are independent attributes of colour.

This plethora of spatial and visual features can be used to encode multiple data attributes. Thus while a graph is, at first sight, two dimensional, in fact it is really multidimensional because of these additional attributes. Some tools or coding systems can give you the flexibility to encode multiple visual attributes like x, y, radius, and colour (e.g., using Google Motion Charts, GoogleVis, D3).

However, while this can be very useful, the resulting graphics can be difficult to understand, it may be difficult to perceive the relationship between different attributes and care must be taken when choosing how to map data attributes on to the spatial and visual features as different features are suited to encoding different kinds of data. Just because something looks good, does not mean that it works well.



Bubble chart with colour (n.d.) Monash University



Interesting case

Chernoff faces

One of the more unusual ways to show multivariate data was devised by Herman Chernoff.

Part of the human visual system is specialised for human face recognition. Having a specialised component for this makes evolutionary sense because it is important to be able to quickly recognise the face of a person approaching with a spear in their hand so that you know whether you should give them a hug or start running.

Chernoff invented the use of little cartoon faces, now called *Chernoff faces*, whose features such as the shape, size of ears, nose, eyes, colour and angle of eyes and mouth can each encode a separate data attribute.

Do you find these easy or difficult to read?

Read more about Chernoff faces in FlowingData's [How to visualize data with cartoonish faces ala Chernoff](#) article (Yau 2010)

Crime by state

The following are examples of some of the states and some of the keys to reading or decoding:

- "height of face" = "murder"
- "height of mouth" = "aggravated_assault"
- "width of mouth" = "burglary"
- "height of eyes" = "motor_vehicle_theft".

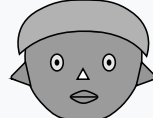
United States



Alabama



Alaska



Arizona



Arkansas



California



Colorado



Connecticut



Delaware



District of Columbia



Florida



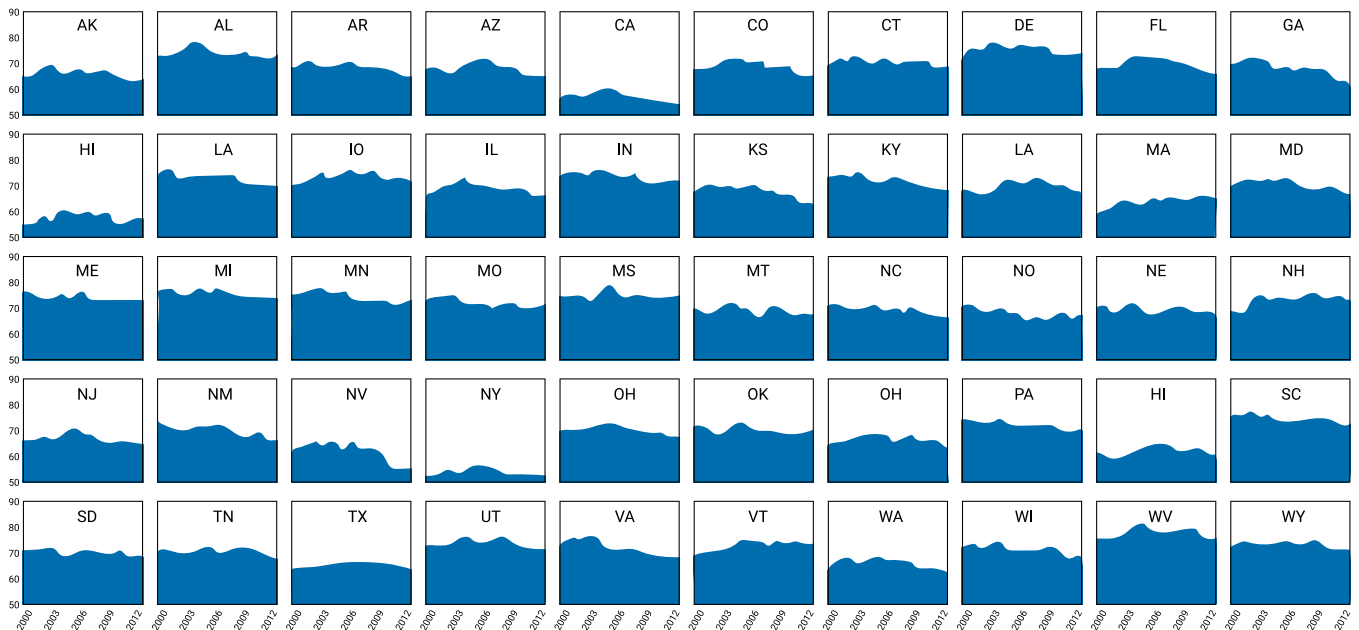
Georgia



Chernoff faces, crime by state (n.d.) Monash University

3.3. Multiple charts

One approach to dealing with multivariate data is simply to use multiple graphics. One of the most common ways is to choose one of the variables and for each value of this variable (or selected values if there are too many) create a chart and then place the resulting charts in a grid, column or row using a common scale so that you can easily compare values between the charts. Tufte calls these **small multiples**. Each chart shows a slice or cross-section through the data.



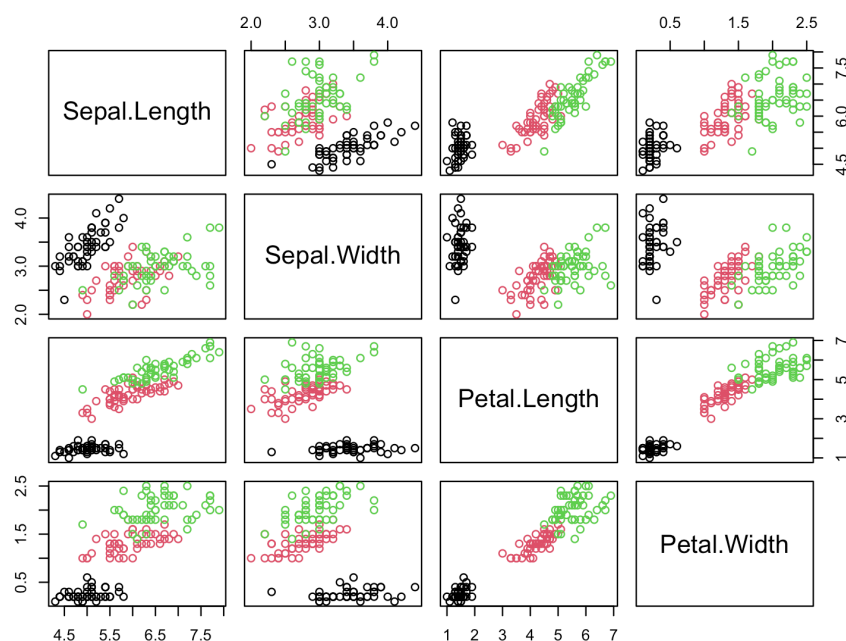
Evolution of home ownership rates across the USA 2000 - 2013 (2022) <<https://bit.ly/3gEI13s>>

Scatter plot matrix (SPLOM)

Another common way is to use a **scatter plot matrix** or **SPLOM**. As its name suggests, this is an array of 2-D scatter plots. There is a row and column for each dimension and the entry for **row** (x) and **column** (y) shows the scatter plot of the (x, y) values.

This kind of chart is sometimes called a **draftsman's display** because it shows orthogonal projections of the multidimensional data that are analogous to the multi-view orthogonal plans consisting of the ground plan, facades and elevations used by architects and engineers.

Edgar Anderson's iris data



Edgar Anderson's iris data (1935), based on the default plot in R - `plot(iris)`

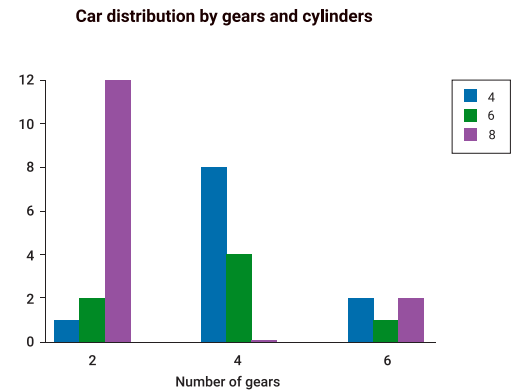
A related approach is to use a **coordinated display** comprising multiple simple charts which may be of different kinds. Each shows a different aspect of the data and these are placed together in a large display. User interaction techniques such as **brushing** and **filtering** allow the viewer to understand how the data attributes in each chart are linked together. For instance, when the user hovers over a point in one chart, all elements corresponding to the same data point will be highlighted in the other charts.

3.4. Overlay graphic elements

Another approach is to create a multi-dimensional graphic by essentially overlaying or juxtaposing the graphic elements from several two-dimensional graphics to form a single chart. This is similar to small multiples but rather than creating a grid of charts, with one chart for each value of some selected variable, the salient graphic elements from these different charts are combined to form a single chart.

Compound and **stacked** bar charts are good examples. Compound bar charts juxtapose the bars from simple bar charts by placing them side by side while stacked bar charts place them on top of each other. And for the record stacked bar charts are better than compound bar charts if comparing the total is important while compound bar charts are better if you wish to compare the size of the different values. Paired bar charts are another special example of a juxtaposed bar chart .

Multidimensional tables, area plots, multi-line graphs, and doughnut charts are the multidimensional equivalents of simple table, line graphs and pie charts. **Stream graphs** are similar to area plots but aren't stacked on a common baseline. Like pie charts, doughnut charts should be avoided.



Compound bar chart (motor cars dataset)
(n.d.) Monash University

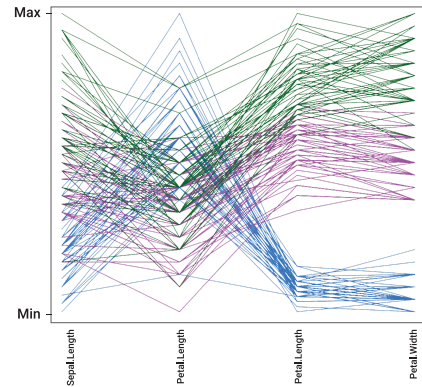
3.5. Parallel coordinates

Yet another approach is to use **parallel coordinates** (see the following graph). Here a sequence of equally spaced vertical lines, one for each dimension, provides the axes for the data. A data point is plotted on these axes by plotting a point on each axis corresponding to the value of the data point's attribute for that dimension and connecting adjacent points by straight lines. Parallel coordinates have a great advantage over the more common Cartesian coordinates that it naturally scales to any number of dimensions.

The crossings of lines between the axes can reveal correlations. If two adjacent attributes are positively correlated then the line segments between them will not cross but if they are negatively correlated they will cross.

One thing to be aware of is that the order of the axes matters, thus when exploring data you need to look at parallel coordinates for different orderings so that, for instance, each pair of dimensions have been placed next to each other in at least one of the plots.

Spider diagrams are a radial version of parallel coordinates. Like most circular plots, I am yet to be convinced of their usefulness.



Parallel coordinates, Iris dataset (n.d.) Monash University

3.6. Dimension reduction

The final approach is to reduce the dimension of the data by projecting the multidimensional data on to two or three dimensions. There are two common ways of doing this.

Multidimensional scaling or MDS

One way is called **Multidimensional scaling** or MDS. Here the idea is to find a location for each data point in 2D space so that points that have similar values for their attributes are close together and those that have dissimilar values are further apart. The visualisation therefore clusters data points that are similar together.

If $\mathbf{x}$ and $\mathbf{y}$ are two data points and $\mathbf{x}_1$ and $\mathbf{y}_1$ are their positions in the 2D plot, then multidimensional scaling tries to make $\text{dist}(\mathbf{x}_1, \mathbf{y}_1)$ as close as possible to $\text{dist}(\mathbf{x}, \mathbf{y})$ where $\text{dist}(\mathbf{u}, \mathbf{v})$ is the distance between two points in either 2D or n -dimensional space. Usually, it is not possible to make these distances exactly equal and so multidimensional scaling will find a position that tries to minimise the overall difference between the distances.

For instance if we have 4 data points with 4 attributes: $[(0,0,0,1), (0,0,1,0), (0,1,0,0), (1,0,0,0)]$ then the Euclidean distance between each pair of points is $\sqrt{2}$ (if we add the distance between corresponding attributes) but there is no way to place 4 points on a piece of paper so that they are equidistant: the best is to place them in a square pattern.

Principal component analysis or PCA

The other common way of reducing dimensions is **principal component analysis** or PCA. This approach finds orthogonal dimensions on which to project the data points that in some sense best explains the variance in the data. More exactly, if you are projecting on to two dimensions PCA first finds the line **L1** for which the variance is greatest when the data is projected on to it and then the line **L2** that is orthogonal to **L1** and which maximises the remaining variance. The data is plotted on a scatter plot where the $\mathbf{x}$ and $\mathbf{y}$ values are the projection of each point on to **L1** and **L2**. While PCA sounds difficult to compute, in fact it is readily computed using an eigenvalue decomposition of the data covariance or correlation matrix. See [wikipedia](#) for more details if you are interested and there's an example of PCA using the Iris data set [here](#).

Self-organising maps (SOMs).

A third approach which is conceptually similar to MDS is **self-organising maps** (SOMs). This is a type of neural network. It is not as widely used as MDS or PCA.

Multidimensional scaling example

In R there is a dataset called 'Eurodist' which is driving distances between 21 European cities (in km, excluding London because you can't drive there and they still measure in miles anyway...).

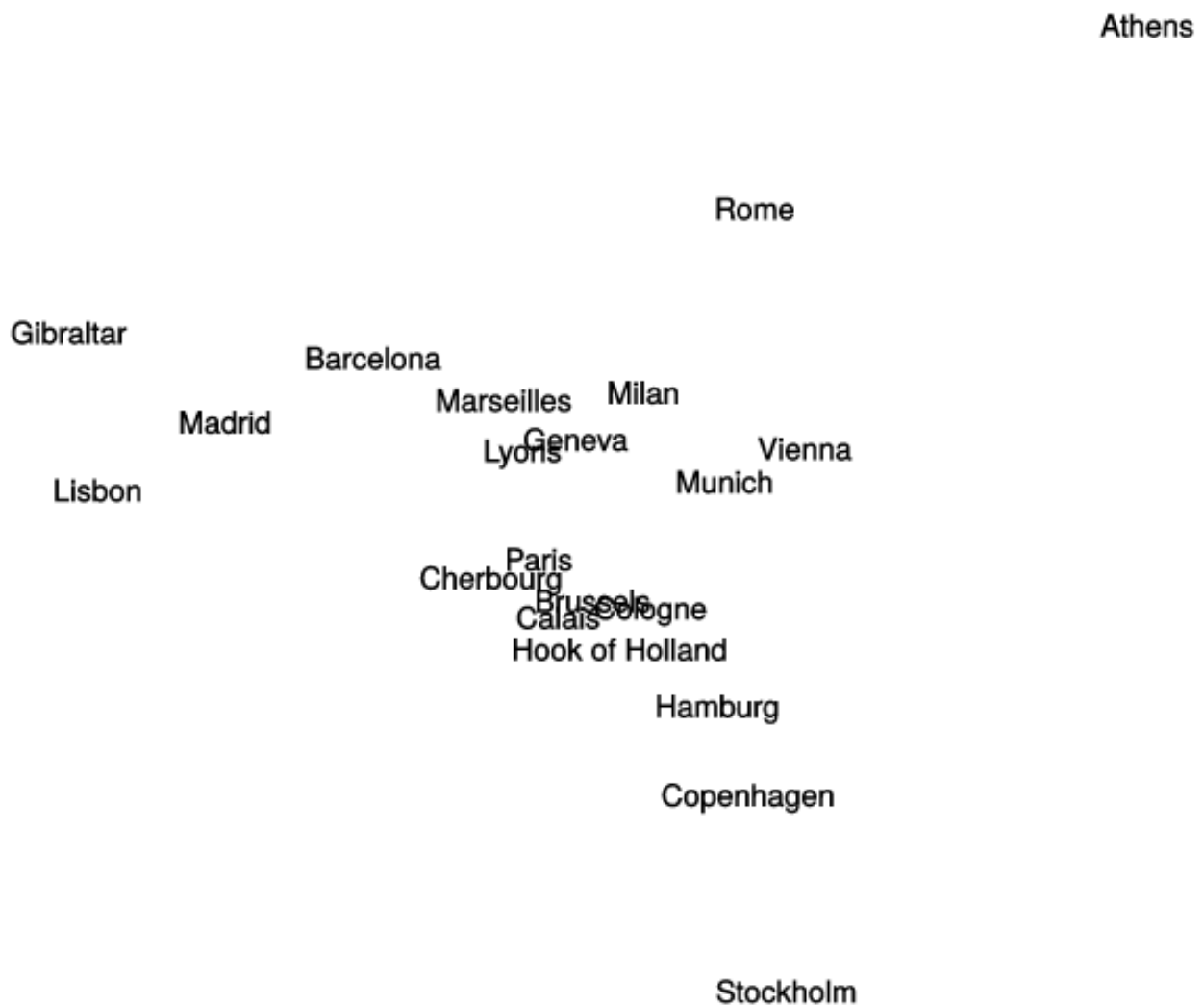
Some of the data is shown in the following table:

	Athens	Barcelona	Brussels	Catala	Cherbourg	Cologne	Copenhagen	Geneva	Gibraltar	Hamburg	...	Lisbon	Lyons	Madrid	Marseilles
Athens	0	3313	2963	3175	3339	2762	3276	2610	4485	2977	...	4532	2753	3940	2865
Barcelona	3313	0	1318	1326	1294	1498	2218	803	1172	2018	...	1305	645	636	521
Brussels	2963	1318	0	204	583	206	966	677	2256	597	...	2084	690	1558	1011
Catala	3175	1326	204	0	460	409	1136	747	2224	714	...	2052	739	1550	1059
Cherbourg	3339	1294	583	460	0	785	1545	853	2047	1115	...	1827	789	1347	1101
Cologne	2762	1498	206	409	785	0	760	1662	2436	460	...	2290	714	1764	1035

Eurodist data table (n.d.) Monash University

After MDS (using 'cmdscale'), we now have a representation of all 21 dimensions of data in a 2-dimensional visualisation (see the following).

cmdscale (stats)



cmdscale(stats) (n.d.) Monash University

When you can compare it with an actual map of Europe (see the following image), Athens (Greece) is a long way (by road) from most of our 21 other European cities, Brussels (for example) is fairly central. Also because MDS doesn't really understand geography, the position is like a distorted mirror of the original positions.



Physical map of Europe (n.d.) <<https://bit.ly/3ODUmSh>>

3.7. Complex graphs

There are many different forms of complex graphs and each has benefits for the way we wish to present our data visually.

In the following video Michael Niemann presents some web resources which are intended to catalogue and clarify the different visualisation types.

These captions have been automatically generated. As such, the quality is not 100% accurate. Accuracy may be affected by audio quality, topic under discussion and clarity of the speakers. Staff are not expected or required to edit or correct automatically-generated captions.

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Multivariate visualisations (2022) Monash University <[Panopto Viewer](#)>

Identifying complex graphs



Investigate

The following web resources explore many different types of data visualisations, explaining the purpose and usefulness of the visualisation types.

- [The Data Visualisation Catalogue](#) (The Data Visualisation Catalogue n.d.).
- [From data to viz](#) (from Data to Viz 2018).
- [Financial Times Visual Vocabulary](#) [GitHub](#) (The Financial Times Limited 2022).

4. Visual variables

When dealing with data and trying to visualise it, it's important to be clear about what the visualization means with respect to the data. That is done by mapping the data items to various elements within the visualisation.

The following table contains eight key visual variables identified by Jaques Bertin in *Semiotique Graphique* (Semiology of Graphics) in 1967.

	Points	Lines	Areas
Position			
Shape			
Size			
Orientation			
Colour hue			
Colour value			
Colour saturation			
Texture/pattern			

Bertin's semiology of graphics (1967) Monash University

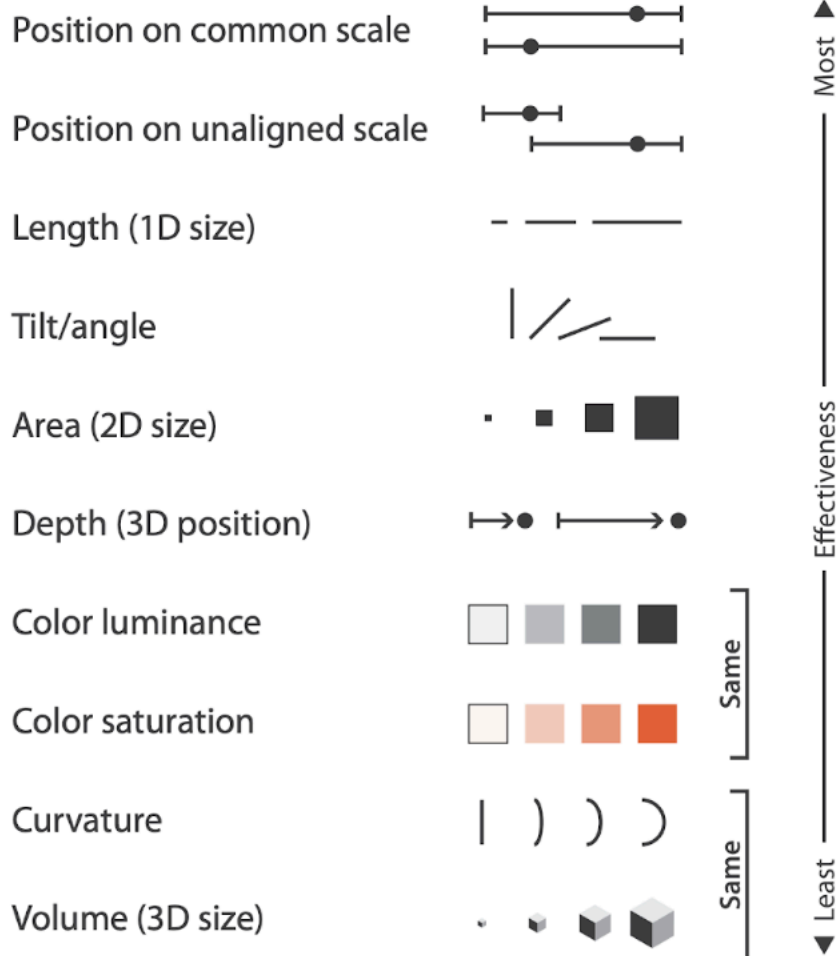
(Text alternative)

Since the 1970s visualisation researchers have tested the effectiveness of visual variables and ranked them based on their effectiveness for different types of data. The original 8 visual variables have also been adapted and renamed as they become better understood.

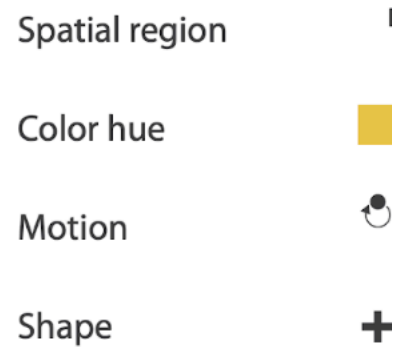
An alternative name for visual variables is visual channels:

Channels: Expressiveness Types and Effectiveness Ranks

➔ Magnitude Channels: Ordered Attributes



➔ Identity Channels: Categorical Attributes



Here you can see for instance size has been split into length, area and volume with length being far more effective in being able to effectively read the data and therefore higher up the ranking. Note which are the most effective and least effective.

This illustration is from *Tamara Munzner's famous textbook "Visualization Analysis and Design"*. (*A K Peters Visualization Series, CRC Press, 2014, with illustrations by Eamonn Maguire*).

Watch the following video in which Dr Michael Niemann describes and evaluates some of the most common visual variables.

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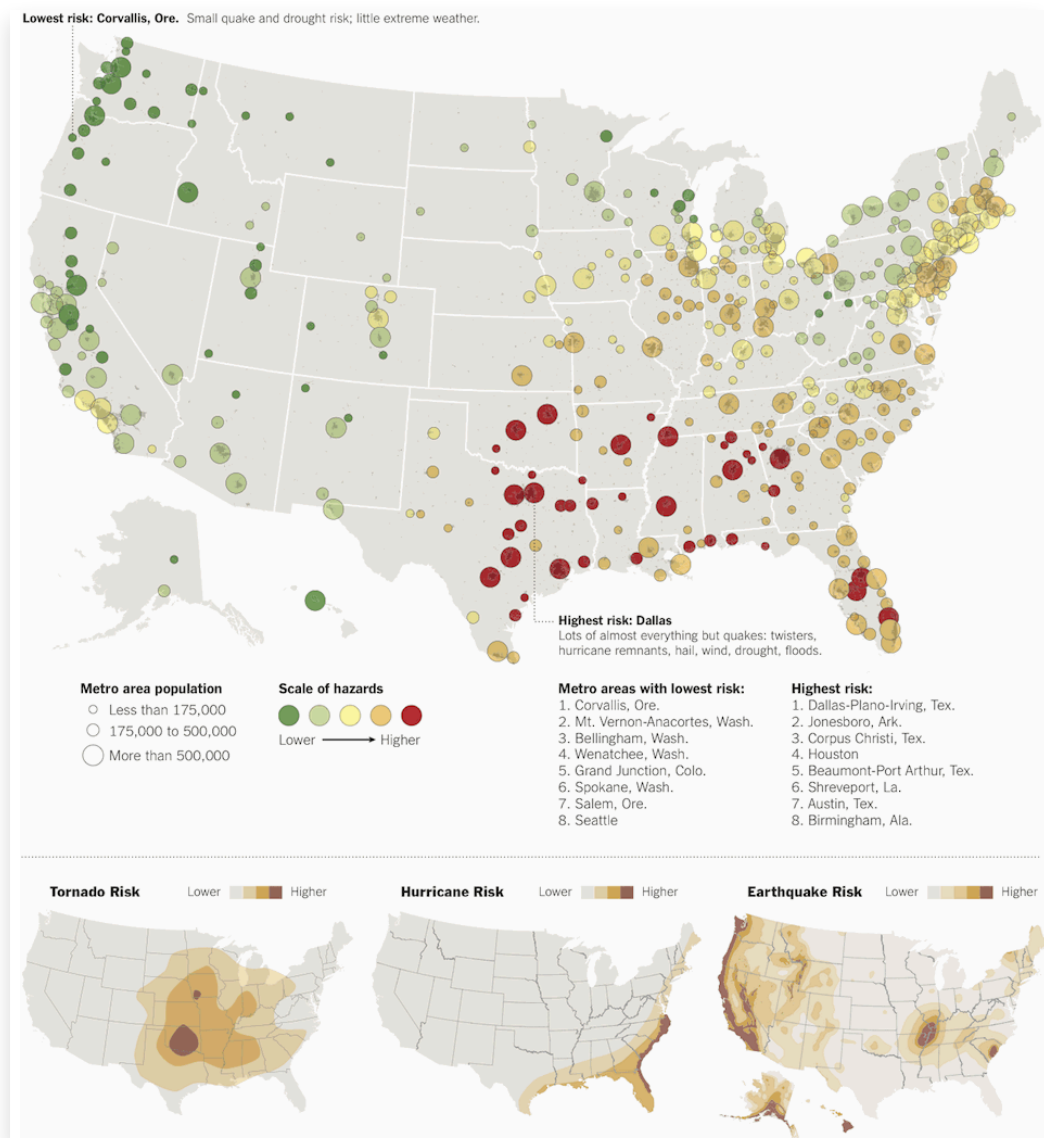


Visual language (2022) Monash University <[Panopto Viewer](#)>

4.1. Analysis of Visual Variables

How are we visually encoding the data?

Consider the following visualisation of data depicting natural disaster risks in the USA and look for the visual variables which have been used to represent data.



Safest places from natural disasters (2011) Sperling's Best Places



Applied learning

Think about how Bertin's Semiology of graphics is being used in this graphic visualisation.

- Which visual variables can you identify being used in this visualisation?
- What attributes of the data is each visual variable presenting?
- Which alternative visual variables do you think could have been used?

Now that you have considered the use of visual variables in the 'Safest places from natural disasters' visualisation, watch the following video, in which Dr Michael Niemann provides an analysis of the example.

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Identifying visual variables (2022) Monash University <[Panopto Viewer](#)>

4.2. Identifying visual variables

Purpose

This activity is intended to check your understanding of visual variables

Task

Step 1: Use your understanding of visual variables to attempt this activity which asks you to identify which visual variables are mapped to the data in a series of example visualisations.

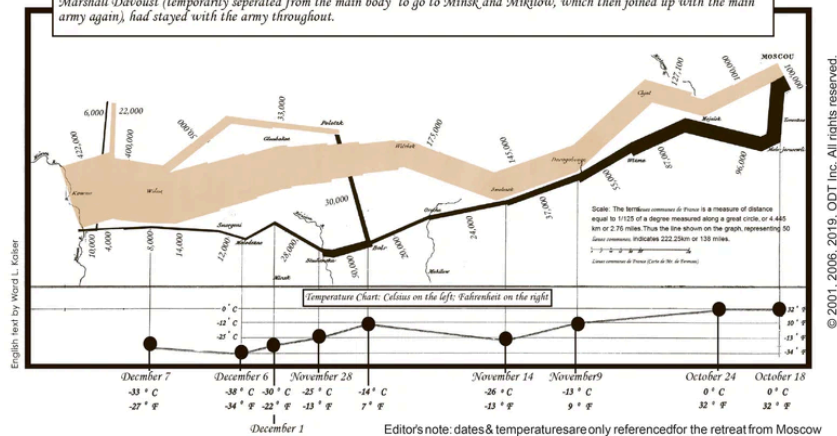
Select each of the options that you think apply to this visualisation. Use the   controls to advance to the next visualisation.

Check your responses using the  Check button.

Minard's Map of Napoleon's March on Moscow

Map representing the losses over time of French army troops during the Russian campaign, 1812-1813.
Constructed by Charles Joseph Minard, Inspector General of Public Works retired.
Paris, 20 November 1869

The number of men present at any given time is represented by the width of the grey line; one mm. indicates ten thousand men. Figures are also written besides the lines. Grey designates men moving into Russia; black, for those leaving. Sources for the data are the works of messrs. Thiers, Segur, Fezensac, Chambray and the unpublished diary of Jacob, who became an Army Pharmacist on 28 October. In order to visualize the army's losses more clearly, I have drawn this as if the units under prince Jerome and Marshall Davoust (temporarily separated from the main body to go to Minsk and Mikhilow, which then joined up with the main army again), had stayed with the army throughout.



Minard's map of Napoleon's Russian campaign.

This graphic has been translated from French to English and modified to most effectively display the temperature data.

Minard's map of Napoleon's march on Moscow (n.d.) <<https://bit.ly/3PmcyjK>>

Identify what visual variables are used (or could be used) in this visualisation.

☐ Position

☐ Shape

☐ Size

☐ Orientation

☐ Colour - hue

☐ Colour - value

☐ Colour - saturation

☐ Texture/pattern

5. End of topic

Well done on reaching the end of **Topic: Tabular data**.



Reflect

In this topic, we have reviewed the basic statistical graphics and then looked at a wide variety of different graphics that have been invented to show multivariate data. While the resulting graphics are invaluable for understanding multidimensional data, they require some skill in interpreting and so must be used carefully.

Next

You may now leave this topic book and return to this week's page. Talk to your peers and instructors if you have any thoughts or questions about the concepts, methods and examples mentioned related to this topic.